

COURSE DESCRIPTION			
Program	Biomedical Engineering, Electronics and Communication, Electrical & Electronics Engineering, Electrical Engineering, Electronics Engineering, Electronics Engineering (Embedded Systems), Electronics and Computer Engineering, Electrical Engineering & Electric Vehicles Technology, Integrated Circuit Design & Fabrication, Instrumentation Engineering, Micro Electronics, Renewable Energy		
Course Title	Mathematics for Electrical Sciences		
Course Code	2001B	Semester	II
Type of Course	Theory	Contact Hours	60
Teaching Scheme (L: T:P:R)	3:1:0:0	Credits	4
CIE Marks	40	ESE Marks	60

Introduction

This course introduces fundamental mathematical tools essential for engineering studies. It covers complex numbers, vector algebra, integral calculus, differential equations and Laplace transforms. These topics are designed to provide the analytical foundation required for solving a wide range of engineering problems, particularly in areas such as system analysis and signal processing. Emphasis is placed on developing skills for solving engineering problems.

Course Objectives

1	To provide comprehensive introductory-level coverage of complex numbers.
2	To familiarize students with the basics of vector algebra.
3	To introduce the fundamentals of integral calculus for modelling and solving engineering-related problems.
4	To solve differential equations and familiarize with Laplace transforms.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Trigonometry, Determinants, Differentiation	1002	Fundamentals of Engineering Mathematics	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Apply concepts of complex numbers to solve problems.	Apply
CO2	Identify and apply vector algebra techniques to solve problems involving scalar and vector quantities in engineering contexts.	Apply
CO3	Use integration techniques to solve basic engineering problems.	Apply
CO4	Solve first-order differential equations and understand Laplace transforms.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-
Total	12	-	-	-	-	-	-	-	-	-	-
Correlation Level	3	-	-	-	-	-	-	-	-	-	-

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme					
							Theory			Practical		
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total

3	1	0	0	5.5	60	4	40	60	100	0	0	0	100
L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination													

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Complex Numbers	The Complex Number System, Fundamental operations with complex numbers, Modulus and Amplitude, Graphical representation of complex numbers, Polar form	11	4
Module II	Vector Algebra	Scalars and vectors, vector addition, subtraction, multiplication by a scalar, Dot product, Cross product, Applications	12	3
Module III	Integral Calculus	Integration as the reverse process of differentiation, Standard results, Integration by parts, Integration by substitution of $f(ax+b)$, $f(x) * f'(x)$ and $f'(x)/f(x)$, Definite integrals, Area bounded by a curve and the x-axis.	11	4
Module IV	Differential Equations and Laplace Transforms	Differential Equations, Laplace transform of elementary functions, Shifting property	11	4

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
The complex number system	Identify the real part and imaginary part	L	CO1	Understand	1
Conjugate of a complex number, Equality of complex numbers	Determine the conjugate of a complex number, Test the equality of complex numbers by comparing their real and imaginary parts.	L	CO1	Apply	1
Fundamental operations with complex numbers : Addition, Subtraction, Multiplication, Division	Perform fundamental operations on complex numbers	L	CO1	Understand	5
Graphical Representation	Represent complex numbers graphically on the Argand (complex) plane.	L	CO1	Understand	1
Absolute value and Amplitude of a Complex Number	Determine the Modulus (Absolute value) and amplitude (argument) of a complex number in degrees	L	CO1	Apply	2
Polar Form	Express a complex number in polar form	L	CO1	Understand	1
Problems	Solve simple engineering problems involving complex numbers.	T	CO1	Apply	4
Module II					
Concept of scalars and vectors, Types of vectors, The unit vectors i,j,k, Position vector of a point, Representation of a vector in terms of i,j,k	Distinguish between scalar and vector quantities and identify components of a vector	L	CO2	Remember	1
Magnitude of a vector	Determine magnitude of a vector and find the unit vector in the direction of a given vector	L	CO2	Remember	1
Addition and subtraction of vectors, Multiplication by a scalar	Perform addition and subtraction of vectors, multiplication of vectors by a scalar	L	CO2	Remember	2
Dot Product	Compute the dot product of vectors and apply it to determine angle between vectors and perpendicularity	L	CO2	Remember	2
Cross Product	Evaluate the cross product of vectors, parallelism	L	CO2	Remember	2
Application of vectors in Engineering : area of parallelogram, area of triangle, work done, moment	Apply vector algebra to solve engineering problems involving area of a parallelogram, area of a triangle, work done, and moment of a force	L	CO2	Remember	4

Problems	Solve and conceptual problems based on vector operations	T	CO2	Remember	3
Module III					
Integration as the reverse process of differentiation	Express integration as the inverse process of differentiation	L	CO3	Remember	1
Integral of standard functions(x^n , e^x , $1/x$, $\sin x$, $\cos x$, $\sec^2 x$, $\operatorname{cosec}^2 x$, $\sec x \tan x$, $\operatorname{cosec} x \cot x$)	Define standard integrals of basic functions	L	CO3	Remember	1
Rules of integration and simple problems, Integration by substitution $f(ax+b)$, $f(x) \cdot f'(x)$ and $f'(x)/f(x)$	Use basic rules of integration (constant multiple, sum and difference rules) to solve simple integrals.	L	CO3	Understand	2
Integration by parts rule and problems	Apply the integration by parts method to evaluate integrals of product of two functions	L	CO3	Apply	2
Definite integrals and simple problems	Evaluate definite integrals	L	CO3	Apply	2
Application of definite integrals	Determine area bounded by a curve and the x-axis	L	CO3	Apply	3
Problems	Evaluate integrals	T	CO3	Apply	4
Module IV					
Differential equations, order and degree of D.E	Identify and classify differential equations by determining their order and degree.	L	CO4	Understand	1
First order differential equations in variable separable form, First order L.D.E, $dy/dx+Py=Q$	Solve first order differential equations (variable separable and linear)	L	CO4	Apply	2
Introduction to Laplace Transforms	Define the Laplace transform of a function.	L	CO4	Remember	1
Laplace transform of elementary functions (1 , t^n , e^{at} , $\sin at$, $\cos at$) and Linearity Property	Evaluate the Laplace transforms of functions	L	CO4	Apply	3
Shifting property (without proof)	State the shifting property of the Laplace transform and solve problems	L	CO4	Apply	2
Inverse Laplace Transform of $1/s^n$, $1/(s-a)$, $1/(s^2+a^2)$, $s/(s^2+a^2)$	Evaluate inverse Laplace transform of $1/s^n$, $1/(s-a)$, $1/(s^2+a^2)$, $s/(s^2+a^2)$	L	CO4	Apply	2
Problems	Solve problems involving Differential equations and Laplace transforms.	T	CO4	Apply	4

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Find conjugate of complex numbers, Determine modulus and amplitude and write polar form of complex numbers	4
2	Solve problems related to equality, addition and subtraction of complex numbers	4
3	Multiply and divide complex numbers	4
4	Find the applications of complex numbers in respective engineering areas.	3
Module II		
5	Identify scalars and vectors, Solve problems on vector addition and subtraction, Multiply vectors by a scalar	4
6	Perform dot product and cross product of vectors	4
7	Solve problems to find work done and moment	4
8	Find the applications of vectors in respective engineering areas.	3
Module III		
9	Find integrals of functions	4
10	Evaluate definite integrals	4
11	Evaluate area bounded by a curve and the x-axis	4
12	Find the applications of integral calculus in respective engineering areas.	3
Module IV		
13	Solve first order differential equation	6

14	Solve problems using first shifting property. Evaluate Laplace transform of functions	6
15	Find the applications of differential equations and Laplace transforms in respective engineering areas.	3
Total Self-Learning hours		60

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Mathematics for Electrical Sciences		SITTTR Kalamassery

REFERENCE			
Sl. No.	Title	Author	Publication
1	Schaum's outline of Complex Variables	Murray R. Spiegel, Seymour Lipschutz, John J. Schiller	Mc Graw Hill
2	Schaum's outline of Vector Analysis	Murray R. Spiegel	Mc Graw Hill
3	Advanced Engineering Mathematics, 10th Ed.	Erwin Kreyszig	Wiley & Sons
4	A Textbook of Engineering Mathematics	N.P. Bali, Manish Goyal	University Science Press
5	Advanced Engineering Mathematics	B. S. Grewal	Mercury Learning & Information

WEB RESOURCES		
Sl. No.	URL	Modules Covered
	NIL	

Assessment Methodology - Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50	50	60
Converted to						20	20	
Marks	5	15				20		60
Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						Total Marks
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	

2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60

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